

REPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTRE DE L'ENSEIGNEMENT SUPERIEUR

COMMISSION NATIONALE D'ORGANISATION DES
EXAMENS ET CONCOURS

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMITTEE FOR THE
ORGANIZATION OF THE NATIONAL EXAMS AND
COMPETITIVE ENTRANCE EXAMINATIONS

National Exam of Higher National Diploma 2026 Session

Specialty/Option: SWE-CSN-DBM

PAPER: Digital Electronics

Duration: 4 Hours

Credit: 7

INSTRUCTIONS:

- **ANSWER ALL QUESTIONS.**
- **CALCULATORS OF ANY TYPE ARE NOT AUTHORIZED!**

SECTION A: NUMBER SYSTEM AND CODES (20 MARKS)

MCQs, for each question, four answers are proposed. Write the letter for the correct answer against the question number. Each question carries 1 mark.

1. **The radix in number system is _____**
 - a) The number digit of the number
 - b) The number of digit allowed in a number system
 - c) The integer part of the number
 - d) the fractional part of the number
2. **Which of the following is not a binary number?**
 - a) 1111
 - b) 0000
 - c) 1010
 - d) 101C
3. **Positive integers must be represented by**
 - a) signed numbers
 - b) unsigned number
 - c) Both A and B
 - d) None of above
4. **A code in which only one bit changes from one code number to the next is:**
 - a) BCD
 - b) Excess-3
 - c) Gray
 - d) ASCII

5. If the decimal number is made of an integer part and fractional part. Then its fractional part binary equivalent is obtained by _____ the fractional part of number continuously by 2.
- Dividing
 - Multiplying
 - Adding
 - Subtracting
6. In the decimal numbering system, what is the MSD?
- The middle digit of a stream of numbers
 - The last digit on the right
 - The rightmost digit to the decimal point of the number
 - The digit with the smallest weight
7. What is the addition of the binary numbers 11011011010 and 010100101?
- 0111001000
 - 1100110110
 - 1110111111
 - 10011010011
8. Binary subtraction of 100101 – 011110 is
- 000111
 - 111000
 - 010101
 - 101010
9. Perform multiplication of the binary numbers: 01001 × 01011
- 001100011
 - 110011100
 - 010100110
 - 101010111
10. Divide the binary numbers: 1010001 ÷ 1000
- 1010.001
 - 1000.001
 - 1100.001
 - 0011.001
11. Which numbering system uses numbers and letters as symbol?
- Decimal

- b) binary
- c) Octal
- d) hexadecimal

12. The octal number $(651.124)_8$ is equivalent to _____

- a) $(1A9.2A)_{16}$
- b) $(1B0.10)_{16}$
- c) $(1A8.A3)_{16}$
- d) $(1B0.B0)_{16}$

13. Convert binary to octal: $(110110001010)_2 = ?$

- a) $(5512)_8$
- b) $(6612)_8$
- c) $(4532)_8$
- d) $(6745)_8$

14. Using a seven digit binary number, what is the largest possible positive decimal number that can be represented?

- a) Decimal 256
- b) Decimal 127
- c) Decimal 68
- d) Decimal 69.

15. The BCD number for decimal 369 is _____.

- a) 1100 1011 1000
- b) 0011 0100 0111
- c) 1100 0110 1010
- d) 0011 0110 1001

16. The hexadecimal equivalent of the following decimal number $(95.5)_{10}$ is

- a) 5F.8
- b) 57.B
- c) F5.8
- d) 58.4

17. The 2's complement of 10010_2 is

- a) 011101
- b) 0101
- c) 10001
- d) 01110

18. The excess-3 code of the number 5_{10} is

- a) 1000
- b) 0110
- c) 0010
- d) 0101

19. A short hand method to convert a binary number to its hexadecimal equivalent is to:

- a) group binary digits in set of three starting from the left and take their hexadecimal equivalent
- b) group binary digits in set of four starting from the left and take their hexadecimal equivalent
- c) group binary digits in set of four starting from the right and take their hexadecimal equivalent
- d) group binary digits in set of three starting from the right and take their hexadecimal equivalent

20. The 1's complement can be easily obtained by using

- a) Comparator
- b) Inverter
- c) Adder
- d) Subtractor

SECTION B: COMPUTER FUNDAMENTAL (80MKS)

I. Hardware (20 Marks)

1) Define the following acronyms:

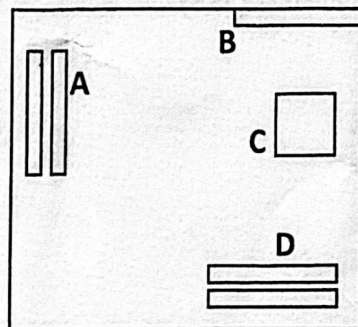
- a) EEPROM,
- b) ROM,
- c) SSD

(3 x 1 = 3 marks)

2)

- a) What is "Cache Memory"? (1 mark)
- b) Give three types of cache memories and their direct functions. (3 marks)
- c) Show a diagrammatic presentation of their positions in a computer system with respect to the processor and the main memory. (3 marks)
- d) Why is it faster than RAM? (2 marks)

3) Below is the basic layout of a motherboard. Label the various parts accordingly.



(a) Label the various parts accordingly. Indicate the components that are connected to these slots (4 marks)

(b) Which form factor is this? (1 marks)

4)

- What is a BIOS?
- Explain three main functions of (a) above (3 marks)

Network And Mobile Devices (40 Marks)

II.

- What is computer network? (1 marks)
- Sketch the topology of each of the following network topology and briefly explain how they function. (3 x 3 = 9 marks)
 - Mesh topology
 - Bus topology
 - Ring topology
- The diagram below is the layout of a LAN in a school.

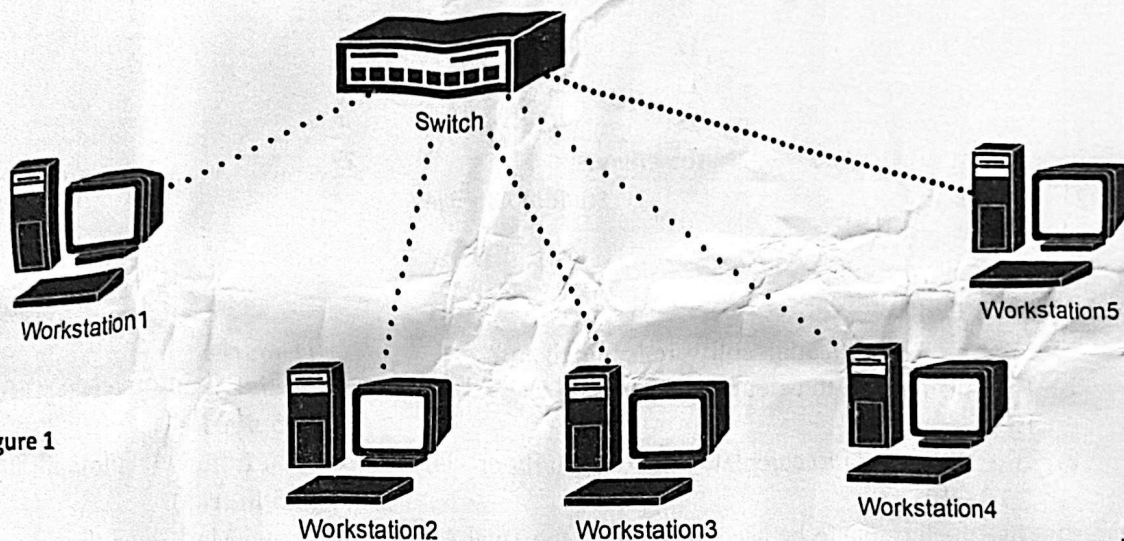


Figure 1

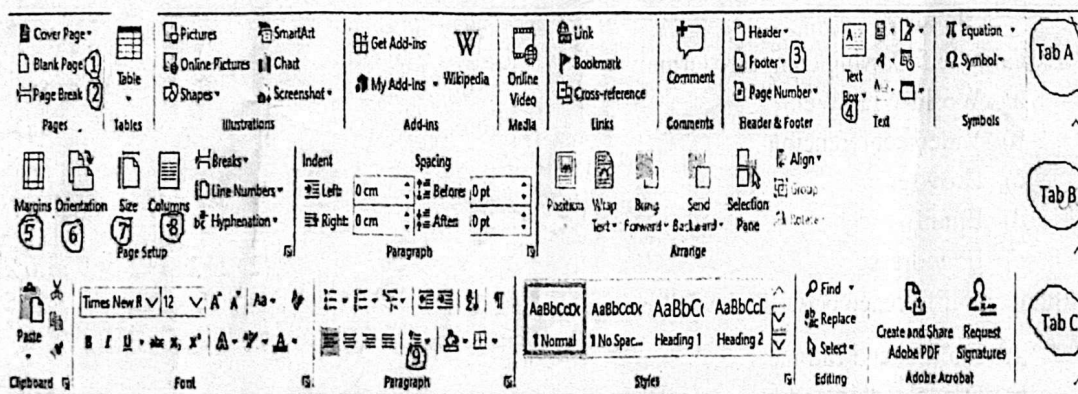
- Identify the type of physical topology used in the LAN of figure 1. (1 mark)
 - What is the role of the switch used in the LAN? (2 marks)
 - What type of network cable is used in this system? Explain your answer. (2 marks)
 - If the service provider gave the address 192.168.100.0/27. Give the IP address of each of the workstation. (5 marks)
- 4) Explain the following computer terms: (2+2+2+1+2 = 9 marks)
- World Wide Web
 - Video conferencing
 - Browser
 - Email
 - IP address
- 2) State one difference between the following: (2 x 2 = 4 marks)
- Internet and Intranet
 - Downloading and Uploading
- 3) Cite three types of mobile devices (3 marks)
- 4) Name five mobile apps and briefly state in what they are useful. (4 marks)

III. Microsoft Word; Excel And Powerpoint (20 Marks)

- 1) What is the suitable office software to perform the following actions?
 a) Write a letter
 b) Write a financial report
 c) Design a calendar
 d) Prepare a presentation for a seminar (4 x 0,5 = 2 marks)
- 2) From the figure below:

	A	B	C	D	E	F	G
1							
2	S/N	Subject	1'st Seq	2's Seq	Average	Coef.	Total Mark
3	1	Maths	15	12	13.5	3	40.5
4	2	English Language	12	8		5	
5	3	French Language	6	14		5	
6	4	ICT	11	15		2	
7	5	Biology	8	9		3	
8	6	History	12	14		3	
9	7	Geography	13	17		2	
10	8	Economic	15	14		4	
11			Total Average			27	
12							
13							

- a) From which application software is this figure? (1 mark)
- b) Give the formula to calculate the average on E4. How can you from E4, automatically fill E5 – E10? (2.5 marks)
- c) Give the formula to calculate the Total Marks on G4. How can you from G4, automatically fill G5 – G10? (2.5 marks)
- d) Give the formula to be used to calculate the Total Average, the Total Marks and the Student Average (3 marks)
- 3) The figure below has four Tab



- a) What is the name of the application? (1.5 marks)
- b) give the name of Tab A, Tab B and Tab C. (3 marks)
- c) give the function of 1, 2, 3 and 4 on Tab A, 5, 6, 7 and 8 on Tab B, 9. (4.5 marks)